Justin Pai

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EDUCATION

The Cooper Union for the Advancement of Science and Art **Expected May 2028**
Bachelor of Science (B.S.), Mechanical Engineering *Manhattan, NY*

- **GPA: 3.6/4.0**; **Relevant Coursework:** Mechanics of Materials, Fluid Dynamics, Thermodynamics, Dynamics, Computer-Aided Engineering (CAD/FEA), Systems Engineering, Differential Equations

WORK EXPERIENCE

L3Harris Technologies **Jun. 2026 – Aug. 2026**
Systems Engineer I (Contract via ProServices Corp) *Clifton, NJ*

- Engineered a **3D** structural ejector seat mounting assembly and support bracket in Creo Parametric to secure the B-52 ejection seat to navigation station support rails.
  - **Key Result:** Applied Design for Manufacturability and Assembly (DFM/DFA) and GD&T principles to optimize multi-axis machining efficiency and achieve **0 tolerance interferences** with existing airframe supports.
  - **Tools & Stack:** Creo Parametric; IBM DOORS; GD&T; Agile; Systems Requirements Traceability
- Authored and refined **10 Contract Data Requirements Lists (CDRLs)**, verifying technical specifications against mechanical and avionics hardware constraints while maintaining **100%** documentation compliance in IBM DOORS.

Cooper Union Summer STEM **Jun. 2025 – Aug. 2025**
Teaching Assistant *Manhattan, NY*

- Mentored **30+ high school students** in mechanical design and mechatronics across an intensive summer program, delivering hands-on instruction in **3 core fabrication technologies**: FDM 3D printing, Onshape, and Arduino.
  - **Key Result:** Coached a **5-student** engineering team through iterative CAD prototyping and tolerance troubleshooting to win **1st place** out of **6+ competing teams** in a judged design competition.

ENGINEERING PROJECTS

Mechanical Rectifier Drive System (EID 103) **Jan. 2026 – May 2026**
Lead Mechanical Designer *Manhattan, NY*

- Designed and fabricated a multi-stage mechanical rectifier assembly to convert oscillating, non-uniform rotational input into continuous unidirectional output.
  - **Key Result:** Reduced output angular velocity fluctuation by **90%** using a custom **4-gear** train, spring-activated lever, and flywheel modeled in Onshape, simulated in MATLAB, and validated via rapid prototyping.

Automated Sun-Tracking Solar Panel (EID 101) **Sep. 2025 – Dec. 2025**
Mechatronics Lead *Manhattan, NY*

- **3D** modeled and fabricated a prototype **2-axis** solar tracking linkage utilizing linear actuators and closed-loop Arduino photoresistor feedback across a **180°** **sweep**, increasing simulated solar capture efficiency by **30%+**.

Geared Stepper Motor Lifting Mechanism (ME 211) **Jan. 2026 – May 2026**
Mechanical Designer *Manhattan, NY*

- Constructed a motorized hoist using a NEMA **17** stepper motor with a custom **6:1** spur gear reduction to multiply torque and reliably lift a **5-lb** payload across an **18-inch** vertical travel.

EXTRACURRICULAR ACTIVITIES

Cooper Union Motorsport (Formula SAE) **Nov. 2024 – Present**
Manufacturing & Powertrain Member *Manhattan, NY*

- Machined **10 custom aluminum** battery accumulator components on a manual vertical mill, interpreting engineering drawings to hold tight tolerances of $\pm$**0.01**″ for precision fitment on the electric race car chassis.
  - **Key Result:** Revised original component geometry in collaboration with powertrain leads to improve cutting tool access, reducing manual mill setup time by **25%** and eliminating material scrap.

TECHNICAL SKILLS

- **CAD & FEA Software:** Creo Parametric; SolidWorks; ANSYS Mechanical (Workbench & APDL); Onshape; AutoCAD
- **Fabrication & Hardware:** Manual Milling; Lathe; FDM 3D Printing (PLA, PETG, TPU); Laser Cutting; GD&T; DFM/DFA; Soldering
- **Programming & Tools:** MATLAB; Simulink; Python; C/C++ (Arduino, ESP32); IBM DOORS; Agile; LaTeX